

Luis Hernández Rocha

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EDUCATION

University of Michigan

August 2026 – Present

Ph.D. in Computer Science

- Advised by Jean-Baptiste Jeannin
- Relevant Coursework: Numerical Linear Algebra, Advanced Programming Languages

Cornell University

August 2022 – December 2025

B.A. in Computer Science, Mathematics with Distinction in all Subjects and Cum Laude in Mathematics

GPA: 4.01/4.3

- Awards: CRA Outstanding Undergraduate Researcher Honorary Mention 2024, 2022-2025 Dean's List
- Relevant Coursework: Category Theory for Computer Science, Graduate Programming Languages, Computer Hardware, Analysis of Algorithms, Operating Systems, Advanced Linear Algebra, Numerical Analysis, Real Analysis, Abstract Algebra.

RELEVANT EXPERIENCE

Research Assistant | Cornell University

June 2025 – December 2025

Professor Andrew Myers

- Explored approaches for designing *Verse*, a statically typed functional language for safe program evolution.
- Examined potential type system formulations based on symmetric coercions to support bidirectional updates between programs.
- Developed OCaml prototypes to test design ideas and refine aspects of the language's static semantics.
- Implemented preliminary components of the parser and type checker using *Menhir* and *ocamllex*.
- Evaluated how subtyping, record projection, and coercions interact, informing later design directions for the project.

Research Assistant | Cornell University

June 2024 – March 2026

Professor Justin Hsu

- Defined translational semantics for converting programs between *NumFuzz*, a language for analyzing numerical programs, and *FPCore*, a standard benchmark suite for floating-point research.
- Implemented an OCaml transpiler that performs these conversions and supports multiple semantics-preserving translation modes.
- Extended *NumFuzz*'s type system to support mixed-precision computation, adding 16-, 32-, and 64-bit rounding operators to model low-precision arithmetic.
- Updated the *NumFuzz* type checker to incorporate these new operators and ensure consistency with the extended type system.
- Define a categorical semantics for new feature which adds kinds for list sizes.

ADDITIONAL EXPERIENCE

Teaching Assistant | Cornell University

August 2023 – Present

Professors Nate Foster, Michael Kim, Walker White

- Assist in teaching five semesters of undergraduate courses, including *Introduction to Computer Science*, *Analysis of Algorithms*, and *Programming Languages and Logics*.
- Hold consulting hours to guide students through course material, programming assignments, and conceptual problem-solving.

Nexus Scholar Researcher | Cornell University

June 2023 – August 2023

Professor Sam Tilsen

- Investigated linguistic evidence for the syntactic hypothesis of unaccusative verb behavior.
- Designed and implemented an online norming study using JavaScript and PCIBex to generate validation data for the main experiment.
- Operated eye-tracking and audio equipment to collect experimental data for statistical analysis.

PRESENTATIONS

BURE Symposium | Cornell University

August 2025

Module Evolution Through Bidirectional Subtyping

BURE Symposium | Cornell University

August 2024

Extensions to NumFuzz

Nexus Scholars Symposium | Cornell University

August 2023

Unaccusativity in Syntactic Planning

EXTRACURRICULAR

Member | Underrepresented Minorities in Computing

September 2022 – December 2024

- Contribute to a supportive community that promotes inclusion and belonging in computing.

- Participate in events and discussions focused on professional growth and shared learning within computer science.

TECHNICAL SKILLS

- **Languages:** OCaml, Rocq, Python, C, Java JavaScript, Racket, LaTeX, English (Fluent), Spanish (Native), French (Basic)
- **Developer Tools:** Git, VS Code, Vim